

Computational assignment help: $H = -\frac{\hbar^2}{2m} \frac{d^2}{dx^2} + \frac{1}{2} m \omega^2 x^2$

1. Write ρ for high temp. $\rightarrow$ Trotter Suzuki /

$$\mathbb{E}[T, V] \neq 0 \rightarrow e^{-B(T+V)} \approx e^{-BV/2} \cdot e^{-BT} \cdot e^{-BV/2}$$

high temp ($B \ll 1$): position space propagator $\langle x | e^{-BT} | x' \rangle$

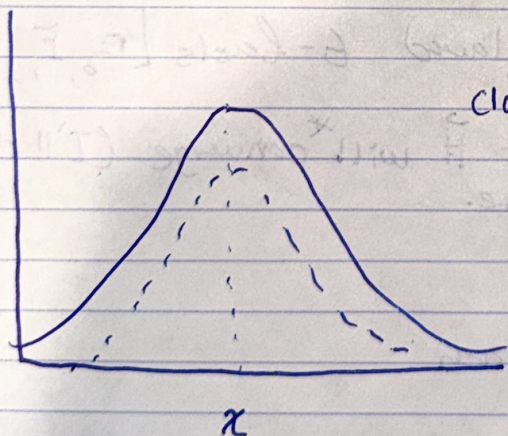
$$\langle x | e^{-BT} | x' \rangle = \sqrt{\frac{1}{2\pi B}} \exp\left(-\frac{(x-x')^2}{2B}\right)$$

and $V(x) = \frac{1}{2} x^2$

(unnormalized) $\hookrightarrow \rho(x, x', B) \approx \sqrt{\frac{1}{2\pi B}} \exp\left(-\frac{(x-x')^2}{2B} - \frac{B}{2} \left(\frac{x^2+x'^2}{2}\right)\right)$

+ for diagonal elements ($x=x'$)

$$\rho(x, x, B) \approx \sqrt{\frac{1}{2\pi B}} \cdot e^{-Bx^2/2}$$



Classical Boltzmann / Gaussian

$\hookrightarrow$ smaller $B \rightarrow$ lower peaked gaussian.

$$B = 0.01$$

$$\rho = \exp(0.01 \left(-\frac{1}{2} \frac{d^2}{dx^2} + \frac{1}{2} x^2\right))$$

Since $\hbar = 1, m = \omega = 1 \rightarrow E_n = (n + \frac{1}{2}) \hbar \omega = n + \frac{1}{2}$

$$\therefore E_0 = \frac{1}{2}$$

2. matrix $p_{ij} \rightarrow$ discretize to x_i

will do via computation later; small $N=3$ grid now.

Let $x = \{-\Delta x, 0, \Delta x\}$, step size Δx ; $x_1 = -\Delta x, x_2 = 0, x_3 = \Delta x$

$\hookrightarrow H = T + V$: + use 2nd order central diff approx

$$\frac{\partial^2}{\partial x^2} \approx \frac{1}{\Delta x^2} \begin{pmatrix} -2 & 1 & 0 \\ 1 & -2 & 1 \\ 0 & 1 & -2 \end{pmatrix} \quad \text{and w/ hamiltonian term} \quad -\frac{1}{2} \frac{\partial^2}{\partial x^2}$$

$$T = \frac{1}{2\Delta x^2} \begin{pmatrix} 2 & -1 & 0 \\ -1 & 2 & -1 \\ 0 & -1 & 2 \end{pmatrix}$$

$V(x) = \frac{1}{2}x^2 \rightarrow$ purely diagonal

$$V_{11} = \frac{1}{2}\Delta x^2$$

$$V_{22} = 0$$

$$V_{33} = \frac{1}{2}\Delta x^2$$

$$\therefore V = \begin{pmatrix} \frac{\Delta x^2}{2} & 0 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & \frac{\Delta x^2}{2} \end{pmatrix}$$

$$H = \begin{bmatrix} \left(\frac{1}{\Delta x^2} + \frac{\Delta x^2}{2}\right) & -\frac{1}{2\Delta x^2} & 0 \\ -\frac{1}{2\Delta x^2} & +\frac{1}{\Delta x^2} & -\frac{1}{2\Delta x^2} \\ 0 & -\frac{1}{2\Delta x^2} & \frac{1}{\Delta x^2} + \frac{\Delta x^2}{2} \end{bmatrix}$$

$$\rho \approx \begin{pmatrix} 1-0.01 \left(\frac{1}{\Delta x^2} + \frac{\Delta x^2}{2}\right) & K & 0 \\ K & 1-0.01 \frac{1}{\Delta x^2} & K \\ 0 & K & 1-0.01 \left(\frac{1}{\Delta x^2} + \frac{\Delta x^2}{2}\right) \end{pmatrix}$$

3. to capture q-behavior grid Δx must be small enough such that quantum behavior reaches neighboring cells.

you used $\rightarrow 0.15$ I went with $\lambda_{th} = \sqrt{\frac{2\pi\hbar^2}{m k_B T}}$

$\hookrightarrow \lambda_{th} = \sqrt{2\pi\beta} \approx 0.25$

So 0.15 or 0.05 scaling works if $[\Delta x \ll \lambda_{th}]$
bc off diag will be resolved across 5x grid points

$p(x, x') \approx C \exp\left(-\frac{(x-x')^2}{2\beta}\right)$ Ratio: $\frac{p(\Delta x)}{p(0)} = \exp(-1.125) \approx 0.325$

(4) high temp Classical: $p_q(x, x) = e^{-\beta V(x)} = e^{-0.00522}$
Quantum: $p_q(x, x) \approx \frac{1}{\sqrt{2\pi\beta}} (e^{-0.0052x^2})$

$Z(\beta) = \text{Tr}(\rho) \Delta x = \Delta x \sum_{i=1}^M p(x_i, x_i)$; high temp $Z(\beta) \approx \frac{1}{\beta}$ if $\hbar = \omega = 1$

$\therefore \text{if } \beta = 0.01 \rightarrow Z \approx 100$

(5) Low temp.: matrix Squaring

as β increases $\text{Tr}(\rho)$ decreases [sys 'occupies' fewer states $\rightarrow e^{-\beta E_0}$ dominates]

if $\beta = 5.12$: classical spikes: $p_q \propto e^{-2.56x^2}$

Quantum: $|\psi_0(x)|^2 = \pi^{-1/2} \cdot e^{-x^2}$

⑥ Extract ground state energy

you did slope of part function [for large β]

grand state contributes: $Z(\beta) = \sum_n e^{-\beta E_n} \propto e^{-\beta E_0}$

↳ plot $\ln(Z)$ v. β ; slope of linear regime

$$E_0 = -\frac{d}{d\beta} \ln(Z) \quad \text{and give } H = -\frac{1}{2} \frac{d^2}{dx^2} + \frac{1}{2} x^2 \rightarrow E_0 = 0.5$$

I would have [possibly inexperience] done direct diagonalization of $\hat{H}$

Matrix squaring finds E_0 thru "imaginary time" evolution
(filters out excited states)

① we fundamentally have eigenvalue problem $[\hat{H} \psi = E \psi]$

② space is discretized then $\hat{H} \rightarrow m \times m$ matrix

③ Eigenvalues of matrix are allowed E -levels $[E_0, E_1, E_2, \dots, E_n]$

④ As $\Delta x \rightarrow 0$, smallest E value of $\hat{H}$ will* converge (I'll check)
to same 0.5 from slope.

Diag. $\rightarrow$ Brute force
matrix squaring $\rightarrow$ stout approach